Ziliang Xiong, Ph.D. Student

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Education

2022 – Now **Ph.D. specialized in computer vision, Linköping University [1], Sweden**

- Research focus: applying uncertainty quantification into computer vision tasks to improve safety and trustworthiness of autonomous systems, including regression, human pose, end-to-end autonomous driving, etc;
- Funded by ELLIIT [2] (Project number: C08), *Situation Aware Perception for Safe Autonomous Robotics Systems*
- WASP [3] affiliation 2023, including PhD courses, summer school, and study visit

2020 – 2022 **M.Sc. Machine Learning, system and control, Lund University [4], Sweden**

- GPA: 4.4/5.0
- Thesis title: *Radar detection using deep learning*.

2015 – 2019 **B.Sc. Automation, Beihang University, China**

- Shenyuan honor's degree

Research Publications

1. **Z. Xiong**, S. Liu, N. Helgesen, J. Johnander, and P.-E. Forssén, "Catplan: Loss-based collision prediction in end-to-end autonomous driving," *under review*, URL: https://arxiv.org/abs/2503.07425.
2. T. Dehdarirad, M. Felsberg, G. Eilertsen, and **Z. Xiong**, "Commutator-induced uncertainty in VAEs," Under review, 2026.
3. S. Liu, **Z. Xiong**, K.-H. Ngo, and P.-E. Forssén, "Matter: Multiscale attention for registration error regression," *2026 IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2026.
4. S. Liu, **Z. Xiong**, B. Wandt, and P.-E. Forssén, "Continuous flow modeling for uncertainty-aware human pose estimation," *Scandinavian Conference on Image Analysis*, 2025.
5. **Z. Xiong**, J. Johanander, and P.-E. Forssén, "Copula calibration for 2d probablistic human pose estimation," Manuscript in preparation, 2025.
6. **Z. Xiong**, A. Jonnarth, A. Eldesokey, J. Johnander, B. Wandt, and P.-E. Forssén, "Hinge-wasserstein: Estimating multimodal aleatoric uncertainty in regression tasks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Jun. 2024, pp. 3471–3480.
7. **Z. Xiong**, S. Kristoffersson Lind, P.-E. Forssén, and V. Krüger, "Uncertainty quantification metrics for deep regression," *Pattern Recognition Letters*, vol. 186, pp. 91–97, 2024, ISSN: 0167-8655. DOI: https://doi.org/10.1016/j.patrec.2024.09.011.

[1] THE Young University Rankings 31

[2] ELLIIT is a strategic research environment in information technology and mobile communications, funded by the Swedish government in 2010. With four partners, Linköping University, Lund University, etc, ELLIIT constitutes a platform for both fundamental and applied research, and for cross-fertilization between disciplines and between academic researchers and industry experts.

[3] The Wallenberg AI, Autonomous Systems and Software Program (WASP) is a major national initiative for strategically motivated basic research, education and faculty recruitment. It is by far the largest individual research program in Sweden.

[4] QS World University Ranking 72

Grants & Projects

2022 ***Situation Aware Perception for Safe Autonomous Robotics Systems*** (ELLIIT Project Co8, Linköping University)

- Applying uncertainty quantification to computer vision tasks to improve the safety and trustworthiness of autonomous systems
- Research areas include regression, human pose estimation, and end-to-end autonomous driving

2026 ***Risk Assessment in Vision-Language-Action Models for Automated Vehicles*** (Vinnova [5] FFI, Linköping University, 240K SEK)

- Developing risk assessment methods for vision-language-action models in automated driving
- Funded by Vinnova (Diarienummer: 2026-00800) under the traffic-safe automation program

2021 **A Vision-based Weed Detection System for the Farmbot** (Lund University, Robotics Lab)

Developed a weed detection system that can aid FarmBot to pick up weeds

- Trained YOLO v3 with Darknet on a customized dataset, reaching 83.2% accuracy on the test set
- Processed and labeled 250+ plant images in 10 classes to build a customized dataset
- Developed software that drives Farmbot to scan the plants, detect and grip the targets
- Calibrated the camera and ed the module to project detections to real-world position

Perspective auto-calibration for security camera applications(Axis Communication A.B.)

- Developed regression modules to calculate perspective, reducing false positive for object detectors in a post-hoc fashion
- Developed a novel visualization module to evaluate ML solutions in qualitative and quantitative ways
- Designed, recorded, and processed videos to form an internal dataset for training and testing regression modules

Employment History

2022. Sep – Now **Master's Thesis Supervisor, Teaching Assistant, Laboratory Assistant, Linköping University**
Supervising master's theses, project courses and hosting exercise

2021. Sep – 2021. Nov **SI-PASS Leader** European Centre for SI-PASS, Lund University
Hosting supplementary instruction sessions for the course, Image Analysis

2021. Jun – 2021. Sep **Machine Learning Developer** R&D department, Axis Communication A.B.
Responsible for the project, perspective auto-calibration for security camera applications

Teaching Experience

2023 – Now Machine Learning for Computer Vision (Master's level); Embedded Perception Systems (Master's level); Signal and System (Bachelor's level), Images and Graphics, Project Course

2021 – 2022 Image Analysis (Master's level)

[5] Vinnova is Sweden's national innovation agency, funded by the Swedish government to support research and innovation. FFI (Strategic Vehicle Research and Innovation) is a joint programme between the government and the automotive industry focusing on safe and sustainable transport.

Mentorship as Primary Advisor

2026	1	Han Xia, **Calibrating Multi-Modal Trajectory Predictions with Pre-rank Regularization: A Case Study on HiVT and Argoverse**, Linköping University
2025	2	Victor Eriksson, **Learned Visibility Filtering for Accurate 2D-to-3D Semantic Projection in LiDAR Point Clouds**, Linköping University
	3	Axel Söderlind and Philip Welin-Berger, **YOLO11-LLD: End-to-end object detection and monocular distance estimation for low-light traffic situations**, Linköping University
	4	Fredrik Hansson and Lina Tunell, **Video Event Identification using Weakly-Supervised Anomaly Detection**, Linköping University
2024	5	Adam Borgstrand, **Confidence Calibrated Point Cloud Segmentation with Limited Data**, Linköping University
	6	Oskar Savinainen, **Uncertainty Estimation and Confidence Calibration in YOLO5Face**, Linköping University
	7	Shipeng Liu, **Uncertainty-Aware Human Pose Estimation Using CNFs**, Linköping University
2023	8	Johannes Hägerlind, **3D-reconstruction of the Common Murre**, Linköping University
	9	Erik Lidman, **Investigation of Practical Aspects of Multi-Camera Perception for Autonomous Vehicles**, Linköping University

Academic Activity

Peer Review — Reviewer for **International Journal of Computer Vision (IJCV)**, **Pattern Recognition Letters** (3 times), **CVPR 2025**, CVPR workshop **SAIAD** (2024-2026), and **AAAI** 2026

Miscellaneous Experience

Certification

2025, July	**International Computer Vision Summer School**, Spatial Intelligence
2024	**Long Oral Certificate**, Safe Artificial Intelligence for All Domains, CVPR workshop
2023	**Pariticipant Certificate**, WASP Summer School Synthesis of Human Communication
2022	**Affiliated WASP PhD Student positions**, Awarded by Wallenberg AI, Autonomous Systems and Software Program
	Master's thesis scholarship, Awarded by Zenseact A.B.
2016 – 2019	**Scholarship for Excellent Study** , Awarded by Beihang University for three years
2017	**Second Prize-China Undergraduate Mathematical Contest in Modeling**. Awarded by China Society for Industrial and Applied Mathematics.

Skills

Languages	Strong reading and writing competencies for English, Mandarin Chinese, Swedish (A1)
Coding	MATLAB, Python, Tensorflow, Pytorch, Git, Julia, SQL, LaTeX, Jax, Apptainer
Misc.	Academic research, teaching, consultation

References

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